

SANJEEVA REDDY S

International Institute of Information Technology Bangalore

☎ +91 8125351715 ✉ sanjeevareddy.s414@gmail.com 🔗 linkedin.com 🌐 github.com

Professional Summary

- S. Sanjeeva Reddy (Gradute Student Member, IEEE) is a graduate researcher specializing in radar and array signal processing. His research focuses on high-resolution spectral estimation, direction-of-arrival (DoA) estimation, sparse and redundant array synthesis, and antenna array optimization. His broader interests include statistical signal processing, covariance-based estimation methods, and algorithm development for advanced radar systems. He holds membership in the IEEE Signal Processing Society (SPS) and the Aerospace and Electronic Systems Society (AESS).

Education

International Institute of Information Technology Bangalore (IIITB) **2025 – 2027 (Expected)**
MS by Research in Radar Signal Processing under Prof. Vinod Veera Reddy *CGPA: 3.55/4.00*

Vellore Institute of Technology, Vellore **2021 – 2025**
B.Tech. Electronics and Communication Engineering *CGPA: 3.43/4.00*

Professional Experience

Indian Institute of Space Science and Technology (IIST) **May 2024 – July 2024**
Summer Research Intern *Trivandrum, India*

- Implemented a real-time image processing system on Xilinx ZCU104 FPGA using Vivado and Vitis.
- Optimized hardware–software data pipelines for image inversion and gray-level processing.
- Worked with High-Level Synthesis (HLS) and hardware–software co-design workflows.

NERTU, Osmania University **Aug 2023 – Dec 2023**
Research Intern *Hyderabad, India*

- Built a complete software-defined FM receiver using MATLAB and RTL-SDR.
- Implemented filtering, demodulation, and audio processing pipelines.
- Produced technical documentation detailing system performance.

Technical Expertise

Core Areas: Radar Signal Processing, Array Processing, Statistical Signal Processing, Sparse Array Synthesis, Antenna Array Optimization, Direction-of-Arrival Estimation, Spectrum Estimation, Radar Data Processing, mmWave Radars, RTL-SDR.

Programming: MATLAB

Research Projects

1. Covariance Fitting Based Spectral Estimation Method [1]

- Developed a high-resolution spectral estimation algorithm that minimizes the Frobenius norm between the estimated covariance matrix (derived from data) and the true covariance matrix (based on the assumed data model). The algorithm is iterative and employs a gradient descent method.
- Validated the method for MST radar atmospheric datasets.

2. Performance Analysis of DoA Estimation Algorithms using Real Time Data from mmWave Radar [2]

- This work focused on Direction of Arrival (DoA) estimation using real-time data collected from the Infineon BGT60TR13C mmWave radar sensor.
- Prominent DoA estimation algorithms, including Conventional Beamforming (CBF), Multiple Signal Classification (MUSIC), and Sparse Parametric Iterative Covariance-based Estimation (SPICE), were implemented and evaluated using the collected data.

3. Design of Two-Fold Redundant Sparse Arrays (TFRSAs) using Exhaustive Search Methods [3]

- Developed a systematic algorithm to discover optimal TFRSAs for $N > 10$, and obtained a new family of near-/sub-optimal TFRSAs that can be completely specified using closed-form expressions (CFEs)
- Extended the catalogue of known optimal RMRA and formulating a sub-optimal RMRA that abides by CFEs.

- Submitted to IEEE Open Access Journal (Under Final Review)

4. Design Aperture-Limited Two-Fold Redundant Sparse Arrays for Physical Size Constrained Scenarios Using Intelligent Exhaustive Search [4]

- We developed a MATLAB/Python pipeline that performs exhaustive search of integer linear arrays meeting a given physical aperture. We believe this is the first attempt to find aperture-constrained TFRSA configurations using the least number of sensors while ensuring two-fold coarray redundancy.
- Extended the catalogue for each aperture. We not only presented the optimal TFRSA geometries but also identified configurations that provide the least mutual coupling.
- Submitted to IEEE SPACE 2026 Conference (Under Review)

5. Array Converter GUI | MATLAB App Designer2024

- Created a GUI tool to convert array notations and analyze DCA and redundancy.

Certifications

- Licensed Amateur Radio Operator (Restricted Grade) – Department of Telecommunications, India
- MATLAB Onramp, Signal Processing Onramp – MathWorks

Leadership & Activities

Student Affairs Council

2025 – Present

Student Representative

IIT Bangalore

- Serve as the representative body for M.S./Ph.D. students and facilitate liaison activities with the institutional administration..

Chairperson

2023 – 2025

VIT Amateur Radio Club

VIT

- Trained 20+ students and helped them obtain Amateur Radio License .
- Successfully conducted antenna building workshop and trained students in home-brewing Yagi-Uda antennas for radio communication

Executive Organizer and Advisory Member

2023 – 2025

Yantra - A Week Long of Technical Innovation

VIT

- Functioned as the key organizer for the annual technical week at VIT, operating under the Office of Students' Welfare.
- Guided and trained 50+ students in fest and event management
- Handled Finance, Purchase, Logistics, Guest Care, Registrations and coordinated with the Administration for the successful conduct of the fest

References

- [1] S. S. Reddy and G. R. Reddy, "Processing of MST Radar Data Using the Covariance Fitting Algorithm," *IEEE Access*, 2025. doi: <https://doi.org/10.1109/ACCESS.2025.3601379>
- [2] S. S. Reddy, M. Kumar, and Valarmathi, "Performance Analysis of DoA Algorithms Using Real-Time mmWave Radar Data," in *Proc. IEEE SENNET*, 2025. doi: <https://doi.org/10.1109/SENNET64220.2025.11136010>
- [3] S. S. Reddy, A. P., and G. R. Reddy, "Discovering Optimal Robust Minimum Redundancy Arrays (RMRAs) through Exhaustive Search and Algebraic Formulation of a New Sub-Optimal RMRA," *IEEE Access*, **Under Review**.
- [4] A. P., S. S. Reddy, and G. R. Reddy, "Aperture-Limited Two-Fold Redundant Sparse Arrays for Physical Size Constrained Scenarios Using Intelligent Exhaustive Search," in *Proc. IEEE SPACE*, 2026, **Under Review**.